Mridul Sahu

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Senior Software Engineer on Google's Core ML Frameworks team, building and optimizing **Orbax**, JAX's distributed checkpointing library. I work the **JAX / XLA** training stack at the sharding, collective, and compiler level — checkpoint I/O, fault tolerance, and large-model save/load. 7+ years building high-performance distributed systems across ML infrastructure, planet-scale backend services, and low-latency trading platforms.

WORK EXPERIENCE

Google — **Oct. 2021 – Present**

Senior Software Engineer, Core ML Frameworks (Nov. 2024 – Present) — *Bangalore, India*

- **Cut Orbax checkpoint blocking-save time 30–50%** (and substantial annualized TPU/GPU savings) by designing an asynchronous directory-creation mechanism with JAX distributed-client coordination and `CommitFuture` abstractions; awarded Google's internal **Perfy** performance award.
- Designed a **sharding-driven checkpoint loader** that reads from the target `jax.sharding` rather than file layout, coalescing each process's shards into byte-range reads. **Eliminated a per-tensor reshard collective and per-tensor XLA recompilation** and cut cluster read amplification to ~1× the checkpoint.
- Cut large-model restart time and **storage read amplification N×→1×** with **Single-Replica Restore + Broadcast** (one replica loads, then broadcasts across devices); hardened fault tolerance via the Emergency Checkpoint Manager's two-stage local-cache + persistent save.
- Architected Orbax's next-generation **Dispatcher API**, abstracting multi-controller and single-controller JAX (incl. colocated Python) backends behind one extensible interface for distributed checkpointing.
- Built the **Knowledge Distillation** module for **Tunix** (Google's JAX-native LLM post-training library): a strategy-pattern framework for logit, feature/attention transfer, and pooling/projection alignment across heterogeneous architectures. Removed per-step XLA recompilation by hoisting `nnx.jit` to one-time setup and fixed TPU HBM OOM on load via CPU host placement.
- Built Orbax's checkpoint **benchmarking and observability framework**: `jax.monitoring` stage timings, XProf traces, per-device HBM usage, compile-cache hit-rate, and TensorStore I/O, with A/B regression baselines, SHA-256 correctness digests, and CI/CD; validated at llama3-8B/70B/405B scale.
- Resolved high-stakes Orbax **performance and stability issues for key enterprise cloud customers**, unblocking large-scale model training and strengthening strategic partnerships.
- Founding member of the **Core ML India** team; mentor engineers across Keras, Keras Hub, JAX 3P, Orbax, and Gemax Prod.

Software Engineer III, Google One (Oct. 2021 – Oct. 2024) — *Bangalore, India*

Storage Quota Service · Apr. 2023 – Oct. 2024

- Designed an **asynchronous processing system** that eliminated database lock contention and sustained **O(100K) write QPS**, unlocking usage-calculation capacity across all Google services.
- Tech-led a Gmail × SQS initiative for an **XX Billion-user** base: built a scalable experimentation framework, contributed to Gmail's codebase to pilot a 30-day storage grace period, and shipped email + in-Gmail warning banners alerting millions to grace-period status.
- Shipped a phased out-of-storage launch impacting **XX Million users/year** with **zero production incidents**, driving substantial Google One signups and upgrades.
- Built a **data-validation and recovery pipeline** (CPU/storage-optimized, proactive alerting, command-driven recovery) safeguarding integrity for **XX Billion users**, plus a precision outage-recovery mechanism restoring usage across Gmail, Photos, and Drive with backfill for newly onboarded services.
- Built real-time **service-level usage-breakdown** infrastructure for Enterprise Customers, eliminating inefficient caching of millions of user-level rows in admin consoles.

Benefits · Oct. 2021 – Apr. 2023

- Built the foundational **benefits-eligibility backend** (cornerstone of a major customer-acquisition feature), improving performance **70% over v1**, plus a reconciliation framework preventing data inconsistencies across partner teams (Photos, YouTube, Workspace, GStore).
- Led multiple benefit launches and international expansions impacting **XX Million users** as central technical liaison across Product Areas; built backend to dynamically split/combine benefits for tailored delivery.
- Fixed critical infrastructure deficiencies affecting millions of users; established alerting/SLO/monitoring standards; owned a **zero-incident on-call migration** of the Benefits and SQS teams from the US to India.

Zopte Technologies (CrispyGames) **Jun. 2018 – Sep. 2021**
Software Engineer *Mumbai, India*

- Engineered a Python **backtesting and live-trading platform** (Bloomberg, Zerodha, NSE-NNF, QuickFIX, custom TimeSeries DB) with an **~80× performance uplift**, plus a distributed backtesting optimizer for strategy-parameter search over a compute cluster.
- Built a low-level **NSE/NFO exchange** library (proprietary protocols, binary structures, security compliance), an **Order Management System (OMS)**, and Tick Collector; maintained an NSE-licensed CTCL application in production at a brokerage.
- Integrated CDSL DP with E-DIS and built a client-onboarding flow (CVLKRA, CKYC, Digio, e-Signing, PDF processing) for a new internet-trading platform, plus a real-time reporting tool for order/trade KPIs.
- Built a **code generator** producing customizable, mockable server- and client-side codebases (up to **60% of project code**), wired to a custom real-time continuous-deployment system.
- Architected a **microservices service mesh** from scratch (Zookeeper, Redis, Nginx, Elasticsearch, MongoDB, MySQL, gRPC/REST) on Kubernetes with GCP and Istio; built pluggable Flutter / Angular Dart UI components.
- Founded and built an online coding-education platform (**Zopte Academy**): streaming/recording (Nginx, HLS) and a Go + Angular Dart student-progress system.
- **R&D:** ML pipelines (Dlib, Tesseract, TensorFlow, OpenCV) in Python/Go for information extraction; an algorithmic-trading platform predicting patterns via classification/regression; prototyped *Pawsome* (Flutter + Go) for community-based stray-pet care.

CrispyGames **Jun. 2017 – Jul. 2017**
Software Engineer, Intern *Mumbai, India*

- Built 2D cross-platform games (Cocos2dx, Android SDK, Swift, C++) and Java-based (Jappy) web admin tools to inspect running applications, deployed on AWS S3 and EC2.

PROJECTS

research.rudrite.com — Interactive Research-Paper Visualizer **2025 – Present**

- Building a site that **visualizes and explains landmark AI/ML research papers** through 120+ interactive, animated explainers spanning training-at-scale parallelism (Megatron, GSPMD, ZeRO), efficient LLM serving (batching, speculation, KV-cache), transformers, and RLHF / reasoning.

explainers.rudrite.com — Interactive Visual Explainers **2026 – Present**

- Interactive visual explainers of system design, distributed systems, and real code — 19 explainers live, from high-level architecture to line-by-line code walks.

PUBLICATIONS

- C. Gaffney, S. Li, D. Ng, **M. Sahu**, et al. — **"Orbax: Distributed Checkpointing with JAX,"** *arXiv:2605.23066* (cs.DC / cs.LG), 2026. A modular, JAX-native checkpointing library demonstrating up to **3.5x faster saving and 2x faster loading** vs. comparable PyTorch checkpointing.
- S. K. Jaiswal, M. Pal, **M. Sahu**, P. Sahu, A. Dev — "EvoCut: A New Generalization of the Albert-Barabási Model for Evolution of Complex Networks," *22nd FRUCT Conference*, Jyväskylä, 2018, pp. 67–72.

EDUCATION

National Institute of Technology, Meghalaya **2014 – 2018**
B.Tech, Computer Science & Engineering — CGPA 8.76/10 *Shillong, India*

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python, Java, Go, C++, Dart, JavaScript; Ruby on Rails, Swift, Bash (familiar)

ML & Performance: JAX, XLA, Orbax, Flax (NNX), Optax, Tunix, TensorStore, Pathways on Cloud; distributed training, sharding, collectives, TPU, profiling (XProf, jax.profiler, jax.monitoring), knowledge distillation, ML systems

Systems & Infrastructure: distributed systems, microservices, system design, API design, low-latency / trading systems (FIX, OMS, exchange protocols), gRPC, REST; Kubernetes, GCP, Docker, Git, Unix/Linux, XPK, Nginx, Minikube, Hugo

Data, Web & Certifications: SQL, MySQL, MongoDB, Elasticsearch, Redis, Cloud Pub/Sub; HTML/CSS, Angular Dart, Flutter, OpenCV. **Certs:** Generative AI with LLMs (DeepLearning.AI), Learning JAX (LinkedIn), Transformer Models & BERT (Google Cloud), Distributed Systems for Practitioners (Educative)